

1.1: Problem Solving:

A conjecture is a statement which has not yet been proven

1.2: Methods of Proof:

Proof by Deduction / Direct Argument:

Using a direct algebraic argument & manipulation to show the validity of a statement.

Proof by Exhaustion:

Ways of proving all possible cases.

Disproof by Counter-example:

A conjecture cannot be true if there is a single counter example.

Proof by Contradiction:

Assume the conjecture is false, and hence show this is mathematically impossible

Example: Show $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$

Suppose $\sqrt{2} \in \mathbb{Q} \Rightarrow \sqrt{2} = \frac{p}{q}$, $\gcd(p,q) = 1$

$\Rightarrow 2q^2 = p^2 \Rightarrow p^2 \equiv 0 \pmod 2 \Rightarrow p \equiv 0 \pmod 2$

$\Rightarrow p = 2k$

$\Rightarrow 2q^2 = 4k^2 \Rightarrow q^2 = 2k^2 \Rightarrow q^2 \equiv 0 \pmod 2 \Rightarrow q \equiv 0 \pmod 2$

$\Rightarrow q \equiv p \pmod 2$

$\Rightarrow \gcd(q,p) \neq 1 \Rightarrow$ Contradiction

$\therefore \sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$